

Carlos Payá – Curriculum Vitae

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Latest update	May 18, 2026		

Personal Profile

I am a PhD candidate at the QUDYMA and Q4Q research groups at ICMC-CSIC, Madrid, under the supervision of Elsa Prada and Ramón Aguado. My work focuses on the theoretical and computational study of topological superconductors realized in hybrid materials, with a specific emphasis on quasi-1D nanowires and their potential applications in quantum computing. While my expertise lies in theoretical and computational physics, I actively collaborate with several experimental groups, providing sophisticated numerical simulations to support their research.

Education

2023-Present	PhD in Condensed Matter Physics - Universidad Autónoma de Madrid, Madrid, Spain
	<i>Tentative completion date</i> February 2027
	<i>Topic</i> Topological superconductors with hybrid materials: full-shell Majorana nanowires
	<i>Supervisors</i> Elsa Prada and Ramón Aguado
2021-2022	Master in Condensed Matter Physics - Universidad Autónoma de Madrid, Madrid, Spain
	<i>Grade</i> 9.45/10
	<i>Master's Thesis</i> Topological phase and Majorana zero modes in full-shell nanowires
	<i>Supervisor</i> Elsa Prada
2017-2021	Bachelor in Physics - Universidad Autónoma de Madrid, Madrid, Spain
	<i>Grade</i> 8.44/10
	<i>Bachelor's Thesis</i> Josephson junctions based on full-shell Majorana nanowires
	<i>Supervisor</i> Elsa Prada

Employment History

March 2023 - Present	Instituto de Ciencia de Materiales de Madrid (ICMM), CSIC, Madrid, Spain
	<i>PhD Candidate, FPI grant</i>
	Quantum Dynamics of Materials (QUDYMA) and Quantum Materials for Quantum Technologies (Q4Q) groups.
April - July 2025	Niels Bohr Institute, University of Copenhagen, Denmark
	<i>Visiting PhD candidate</i>
	Karsten Flensberg's group.
Nov 2021 - Sep 2022	Instituto de Ciencia de Materiales de Madrid (ICMM), CSIC, Madrid, Spain
	<i>Research Assistant</i>
	Theory of Quantum Materials and Solid State Quantum Technologies group.

Publications

Preprints

- [P9] **C. Payá**, C. Robles, P. San-Jose, and E. Prada. *Absence of Quasi-Majorana False Positives in Full-Shell Hybrid Nanowires*, (2026).

Journals

* indicates first theory author.

- [P8] **C. Payá**, O. Solow, E. Prada, R. Aguado, and K. Flensberg. *Non-Hermitian skin effect and electronic nonlocal transport*, *Physical Review B*, 113, L161405 (2026). Citations: 1.
- [P7] **C. Payá**, F. J. Matute-Cañadas, A. L. Yeyati, R. Aguado, P. San-Jose, and E. Prada. *Fluxoid valve effect in full-shell nanowire Josephson junctions*, *Physical Review B*, 112, 134520 (2025). Citations: 3.
- [P6] **C. Payá**, R. Aguado, P. San-Jose, and E. Prada. *Josephson effect and critical currents in trivial and topological full-shell hybrid nanowires*, *Physical Review B*, 111, 235420 (2025). Citations: 5.
- [P5] M. T. Deng, **C. Payá***, P. San-Jose, E. Prada, C. M. Marcus, and S. Vaitiekėnas. *Caroli-de Gennes–Matricon Analogs in Full-Shell Hybrid Nanowires*, *Physical Review Letters*, 134, 206302 (2025). Citations: 7.
- [P4] A. Vezzosi, **C. Payá**, P. Wójcik, A. Bertoni, G. Goldoni, E. Prada, and S. D. Escribano. *InP/GaSb core-shell nanowires: A novel hole-based platform with strong spin-orbit coupling for full-shell hybrid devices*, *SciPost Physics*, 18, 069 (2025). Citations: 4.
- [P3] **C. Payá**, P. San-Jose, C. J. S. Martínez, R. Aguado, and E. Prada. *Absence of Majorana oscillations in finite-length full-shell hybrid nanowires*, *Physical Review B*, 110, 115417 (2024). Citations: 8.
- [P2] **C. Payá**, S. D. Escribano, A. Vezzosi, F. Peñaranda, R. Aguado, P. San-Jose, and E. Prada. *Phenomenology of Majorana zero modes in full-shell hybrid nanowires*, *Physical Review B*, 109, 115428 (2024). Citations: 21.
- [P1] P. San-Jose, **C. Payá**, C. M. Marcus, S. Vaitiekėnas, and E. Prada. *Theory of Caroli–de Gennes–Matricon analogs in full-shell hybrid nanowires*, *Physical Review B*, 107, 155423 (2023). Citations: 25.

Conferences

Tutorials

- [C19] **Feb. 2026**. “International Workshop on Superconductor Semiconductor Hybrids” (Institut Néel, CNRS, Villard-de-Lans, France). Tutorial title: *Realistic Modeling of 1D Hybrid Superconductors*.

Contributed Talks

- [C18] **July 2025**. “QTYR25” (PhD and Young Scientists in Quantum Technologies Network (PYSQT), Madrid, Spain). Contributed title: *Fluxoid valve effect in full-shell nanowire Josephson junctions*.
- [C17] **Mar. 2025**. “APS Global Physics Summit 2025” (American Physical Society, Anaheim, CA, US). Contributed title: *Josephson effect and critical currents in topological full-shell hybrid nanowires*.
- [C16] **July 2024**. “QTYR24” (PhD and Young Scientists in Quantum Technologies Network (PYSQT), Madrid, Spain). Contributed title: *Phenomenology of Majorana zero modes in full-shell hybrid nanowires*.
- [C15] **May 2024**. “Condensed Matter PhD Program Annual Meeting” (Facultad de Ciencias, Universidad Autónoma de Madrid (UAM), Madrid, Spain). Contributed title: *Phenomenology of Majorana zero modes in full-shell hybrid nanowires*.

Poster contributions

- [C14] **Feb. 2026**. “International Workshop on Superconductor Semiconductor Hybrids” (Institut Néel, CNRS, Villard-de-Lans, France). Poster title: *Non-Hermitian Skin Effect and Electronic Nonlocal Transport*.

- [C13] **July 2025.** “Quantum Designer 2025” (Donostia International Physics Center (DIPC), San Sebastián, Spain). Poster title: *Josephson effect in topological full-shell hybrid nanowires*.
- [C12] **June 2025.** “Workshop on Superconductor-Semiconductor Hybrids” (Niels Bohr Institute, University of Copenhagen, Copenhagen, Denmark). Poster title: *Josephson effect in topological full-shell hybrid nanowires*.
- [C11] **Apr. 2025.** “GRC on Hybrid Superconductor-Semiconductor Devices” (Gordon Research Conferences, Les Diablerets, Switzerland). Poster title: *Josephson effect in topological full-shell hybrid nanowires*.
- [C10] **July 2024.** “Quantum Designer 2024” (Donostia International Physics Center (DIPC), San Sebastián, Spain). Poster title: *Phenomenology of Majorana zero modes in full-shell hybrid nanowires*.
- [C9] **May 2024.** “Quantum matter for Quantum Technologies Workshop” (SPICE, Mainz, Germany). Poster title: *Phenomenology of Majorana zero modes in full-shell hybrid nanowires*.
- [C8] **Apr. 2024.** “European School on Superconductivity and Magnetism in Quantum Materials” (SuperQmap COST action, Gandía, Spain). Poster title: *Phenomenology of Majorana zero modes in full-shell hybrid nanowires*.
- [C7] **Sept. 2023.** “Emergence of Quantum Phases in Novel Materials” (Instituto de Ciencia de Materiales de Madrid (ICMM), CSIC, Madrid, Spain). Poster title: *Majorana zero modes in full-shell hybrid nanowires*.
- [C6] **June 2023.** “Bound States in Superconducting Nanodevices” (TopSquad and AndQC collaborations, Budapest, Hungary). Poster title: *Theory of Caroli-de Gennes-Matricon analogs in full-shell hybrid nanowires*.
- [C5] **May 2023.** “QuantumMatter 2023” (Phantoms Foundation, Madrid, Spain). Poster title: *Theory of Caroli-de Gennes-Matricon analogs in full-shell hybrid nanowires*.
- [C4] **May 2023.** “YouMat2023” (Instituto de Ciencia de Materiales de Madrid (ICMM), CSIC, Madrid, Spain). Poster title: *Theory of Caroli-de Gennes-Matricon analogs in full-shell hybrid nanowires*.
- [C3] **July 2022.** “Frontiers in Condensed Matter Physics” (Niels Bohr Institute, University of Copenhagen, Copenhagen, Denmark). Poster title: *Theory of Caroli-de Gennes-Matricon analogs in full-shell hybrid nanowires*.

Attended

- [C2] **Mar. 2024.** “Workshop on Superconductor-Semiconductor Hybrids” (Niels Bohr Institute, University of Copenhagen, Copenhagen, Denmark).
- [C1] **Sept. 2021.** “Emergence of Quantum Phases in Novel Materials” (Instituto de Ciencia de Materiales de Madrid (ICMM), CSIC, Madrid, Spain).

Teaching

Supervised students

Sep 2024 - César Robles - Bachelor’s Thesis

June 2025 Main supervisor: Elsa Prada

Title: *Quasi-Majoranas in inhomogeneous full-shell hybrid nanowires*

Outreach

March 2023 - ICMM Superconductivity Outreach Team

Present Lecturer and workshopper

Regular activities: monthly talks and demonstrations for high-school students.

Events

- [O9] **Mar. 2026.** “Feria Madrid es Ciencia 2026 (Madrid Science Fair)” (Comunidad de Madrid, Madrid, Spain). Role: *workshopper*.
- [O8] **Sept. 2025.** “European Researchers’ Night 2025” (CSIC, Madrid, Spain). Role: *workshopper*.
- [O7] **Mar. 2025.** “Feria Madrid es Ciencia 2025 (Madrid Science Fair)” (Comunidad de Madrid, Madrid, Spain). Role: *workshopper*.
- [O6] **Sept. 2024.** “European Researchers’ Night 2024” (CSIC, Madrid, Spain). Role: *workshopper*.
- [O5] **Mar. 2024.** “Feria Madrid es Ciencia 2024 (Madrid Science Fair)” (Comunidad de Madrid, Madrid, Spain). Role: *workshopper*.
- [O4] **Nov. 2023.** “Semana de la Ciencia (Science Week)” (CSIC, Madrid, Spain). Role: *workshopper*.
- [O3] **Sept. 2023.** “European Researchers’ Night 2023” (Instituto de Ciencia de Materiales de Madrid (ICMM), CSIC, Madrid, Spain). Role: *workshopper*.
- [O2] **June 2023.** “Ciencia en la Calle” (Casa de la Ciencia, Ciudad Real, Spain). Role: *workshopper*.
- [O1] **Sept. 2022.** “European Researchers’ Night 2022” (Instituto de Ciencia de Materiales de Madrid (ICMM), CSIC, Madrid, Spain). Role: *logistics*.

Funding

Student grants

- [F3] *PhD Fellowship* at ICMM (CSIC) (AEI, PRE2022-101362 for the period 2023-2027).

Participation in funded projects

- [F2] *Correlations, Superconductivity and Topology in Quantum Materials and Technologies* at ICMM (CSIC) (AEI, PID2021-125343NB-I00 for the period 2022-2025). Principal investigators: Ramón Aguado and Elena Bascones.
- [F1] *Topology and Correlations in Quantum Materials and Solid State Quantum Technologies* at ICMM (CSIC) (AEI, PGC2018-097018-B-I00 for the period 2021-2022). Principal investigators: María José Calderón and Ramón Aguado.

Awards

- [A4] *Max Mazín Award*. Max Mazín Foundation and CEIM Foundation, 2021. Received during the period 2018-2021. Awarded each year of my undergraduate studies, 5th to 8th editions.
- [A3] *GEFES Research Award for Students*. Condensed Matter Physics Division, Spanish Royal Society of Physics (GEFES-RSEF), 2020. For the work entitled *Josephson Junctions in Full-shell Majorana Nanowires*.
- [A2] *Excellence Fellowship*. Comunidad de Madrid, 2018.
- [A1] *Premio Extraordinario de Bachillerato*. Comunidad de Madrid, 2017. Top 10 academic records of the region.

Skills

Languages

Spanish	Native	
English	Fluent	C1 Advanced certification
French	Fluent	DALF C1 certification

Programming

OS

Programming Languages

Web Development

Scientific computing

Miscellaneous

MacOs, Linux (Debian and Ubuntu), Windows

Python (advanced), Julia (advanced), C/C++ (basic), MySQL (basic)

HTML5, CSS, JavaScript (basic)

Mathematica (advanced), Quantica (Julia package, advanced), Numpy, Scipy

Makie (Julia package, advanced), matplotlib, Git, $\text{\LaTeX}$, Office